College for Professional Studies
3333 Regis Boulevard
Denver, CO 80221

Regis University
College for Professional Studies
School of Computer & Information Sciences

COURSE TITLE: Operating Systems

COURSE NUMBER: CS 430

COURSE DESCRIPTION:

Studies the organization and operation of computer systems. Includes batch processing, inter-active processing, multiprogramming systems, storage management, data sharing in main storage, resource control, file systems and processor scheduling.

PREREQUISITE COURSES:

CS372, Advanced programming and algorithms

COURSE OUTCOMES:

Upon completion of this course, learners should be able to:

1. Identify and list the goals and major elements of an operating system.
2. Explain the implications of concurrent processing on an operating system.
3. Define and explain the difference between buffering and spooling.
4. Define multi-programming and explain how it is accomplished.
5. Evaluate the need for protection and describe where and how it can be accomplished.
6. Describe the implications of CPU scheduling and evaluate the performance of alternative algorithms for scheduling different types of applications.
7. Define deadlock; list methods to identify system deadlock; and identify and distinguish between the pros and cons of deadlock prevention, avoidance, and detection.
8. Identify the purposes behind, and distinguish between methods of, memory management.
9. Explain virtual memory and its relationship to paging.
10. Describe methods of secondary file management and the effects on the performance of the operating system.
11. Explain the interaction between the operating system and the structure used for file systems.
12. List the implications of distributed hardware on the operating system.
13. Evaluate the tradeoffs in placing the burden of supporting system functions in hardware, the operating system, the application software, and on the user.
14. Describe the basics of an operating system and determine the environment in which this operating system would best perform.

COURSE MATERIALS:

Required Texts:

Silberschatz, A., Galvin, P., & Gagne, G. (2013). *Operating System Concepts*. (9th edition). New York, NY: Wiley & Sons. ISBN13: 978-1-118-06333-0 or ISBN10:1-118-06333-3.

Technology Tools:

1. A PC-compatible computer system running Windows XP, Vista, Windows 7 or Windows 8.
2. Microsoft Office or similar suite
3. Microsoft PowerPoint® (part of Microsoft Office) or the free PowerPoint Viewer

As with most of Regis learning activities, using various software applications to accomplish assignments requires students to exercise a great deal of responsibility for learning how to successfully operate the software applications. There are, however, many quality books on the market that support novice users if you need them.

PRE-ASSIGNMENT:

1. **Read** the following pages in your text, *Operating System Concepts*:

   Chapters 1 and 2

2. Be prepared to ask questions on unclear areas and to respond to questions about information in the assigned reading.

RELATED STANDARDS

Program Outcomes	
	This course enables students to achieve the following Program outcomes by the time of graduation:
	An ability to apply knowledge of computing and mathematics appropriate to the discipline
	An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution
	An ability to design, implement and evaluate a computer-based system, process, component or program to meet desired needs
	An understanding of professional, ethical, legal, security and social issues and responsibilities
	An ability to communicate effectively with a range of audiences
	An ability to analyze the local and global impact of computing on individuals, organizations, and society
	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices

Regis Nine Educational Outcomes	
	Content Knowledge in Discipline
	Diverse Cultures, Perspectives, and Belief Systems
	Arts, Sciences, and Humanities
	Critical Thinking
	Communication
	Technology
	Ethical and Social Responsibility
	Leadership in Service to Others
	Lifelong Learning

Chapter Exercises/Homework Assignments

There are eight weekly Chapter exercises / homework assignments. Each assignment will involve writing a detailed answer to a number of assigned problems, using the concepts discussed in the book and class, and may also include submission of pseudo code, charts, and/or diagrams.

Exams

There are four quizzes. Quizzes will be timed; they might be open book, open notes, closed book, and/or closed notes. Quiz questions will be cumulative, taken from reading assignments and class presentations.

Final project

Your goal in this course is to learn the basics of operating systems for centralized computers in general, and to learn to evaluate the appropriateness of an operating system for a particular environment -- not to learn the details of a particular operating system. To provide a general survey of operating systems, you will perform an in-depth analysis of an operating system as your Final Project.

You will submit a final project by the end of week seven of the course. Oral presentations (classroom only) will be conducted on the last night of the class.

Discussion / Participation

Because of the accelerated nature of the course, class discussion and participation is very important. Class participation/effort is important because we can all learn from each other. Your discussion and participation points can make a difference in the final grade. If you do not participate during any given week, you will lose the discussion/participation points for that week.

Participation means:

1. a. Present in class every session (classroom)
   b. Active participation in the discussion forum(s) every week (online)
2. a. Effectively responds to questions from the facilitator (classroom)
   b. Regularly checks forums and post all required assignments/discussion questions/ items by the deadlines (online)

Late Assignments

If you have not negotiated with the facilitator, assignments turned in late will be graded, then reduced by 15%. Assignments will not be accepted later than one week after due date.



COURSE ASSIGNMENTS AND ACTIVITIES:

Week	Topics	Readings	Graded Assignments or Assessments and Associated Points
1	1. Operating Systems Overview 2. Operating System Components & Services	Ch 1 & 2	Homework #1 3.125% Forum/ Participation 3.125% Final Project top two selections
2	3. Process Concepts 4. Threads 5. Job and Processor Scheduling	Ch 3, 4 & 6	Homework #2 3.125% Quiz 1 6.25% Forum/ Participation 3.125%
3	6. Concurrent Process Synchronization 7. Process Deadlocks	Ch 5 & 7	Homework #3 3.125% Forum/ Participation 3.125% Final Project Outline
4	8. Memory Organization and Management 9. Managing Virtual Memory	Ch 8 & 9	Homework #4 3.125% Quiz 2 6.25% Forum/ Participation 3.125%
5	10. File System Concepts 11. File System Implementation	Ch 11 & 12	Homework #5 3.125% Forum/ Participation 3.125%
6	12. I/O Subsystems 13. Secondary Storage Organization & Mgmt	Ch 10 & 13	Homework #6 3.125% Quiz 3 6.25% Forum/ Participation 3.125%
7	14. Protection Concepts 15. Operating System Security	Ch 14 & 15	Homework #7 3.125% Forum/ Participation 3.125% Final Project 25 %
8	16. Virtual Machines 17. Distributed Systems	Ch 16 & 17	Homework # 8 3.125% Quiz 4 6.25% Forum/ Participation 3.125 % Final Project Oral Presetantion (classroom only)
			PERCENT 100

*__Note to Classroom sections only:__ Exact dates for reading assignments and homework assignments may be one week earlier or later than indicated in the Course Assignments and Activities Grid. Your facilitator's syllabus, handed out the first night of class, will indicate any changes.



COLLEGE FOR PROFESSIONAL STUDIES GRADING SCALE:

Grading Scale and Equivalent Points			
Letter Grade	*Percentage*	*Grade Point*	***Minimum Course Equivalent Percentage*
A	93 to 100	4.00	93
A–	90 to less than 93	3.67	90
B+	87 to less than 90	3.33	87
B	83 to less than 87	3.00	83
B–	80 to less than 83	2.67	80
C+	77 to less than 80	2.33	77
C	73 to less than 77	2.00	73
C–	70 to less than 73	1.67	70
D+	67 to less than 70	1.33	67
D	63 to less than 67	1.00	63
D–	60 to less than 63	.67	60
F	Less than 60	0	Less than 60

*See Maximum Percentages from Course Assignments and Activities Table.

COLLEGE FOR PROFESSIONAL STUDIES POLICIES (LINKS):

You will be asked to enter a valid Regis University UserID and password to access the following policies.

Academic Integrity
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/Academic%20Integrity%20Policy%20and%20Board%20-%20December%2020,%202010.pdf

Attendance Participation
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Attendance%20Participation.pdf

Confidential Proprietary Information Policy
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Confidential%20Proprietary%20Information%20Policy.pdf

Dayton Memorial Library
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Dayton%20Memorial%20Library.pdf

Diversity
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Diversity%20at%20RU.pdf

Equal Access and Disability Services
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Equal%20Access_Disability%20Services.pdf

Human Subjects Review (IRB)
http://www.regis.edu/regis.asp?sctn=ars&p1=agr&p2=irb

Inclement Weather and Class Cancellation
http://www.regis.edu/regis.asp?sctn=StuHandbook&p1=NavCampus&p2=Resc

Late or Incomplete Course Procedures
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Late%20Assignments_Assignment%20Revisions%20Policy%20CPS.pdf

Learner Conduct
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Learner%20Conduct.pdf

Writing Assistance
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Writing%20Assistance.pdf

OTHER INFORMATION:

NOTE TO LEARNERS: On occasion, the course facilitator may, at his or her discretion, alter the Learning Activities shown in this Syllabus. The alteration of Learning Activities may not, in any way, change the Learner Outcomes or the grading scale for this course as contained in this syllabus. Examples of circumstances that could justify alterations in Learning Activities could include number of learners in the course; compelling current events; special facilitator experience or expertise; or unanticipated disruptions to class session schedule.